

ECE332 Exam 1 — Full Answer Key

Fall 2025 · Step-by-Step with Supplement References

Problem 1 (20 pts) — Transformer EMF, Triangular Loop

Given: Equilateral triangular loop, N turns, side a , in y - z plane with normal $d\mathbf{s} = \hat{x} ds$. Magnetic field $\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t)$.

(a) Flux through one turn. (6 pts)

[Suppl: p.1 col 1: SURFACE ELEMENT ds ; DOT PRODUCTS; p.1 col 3: EMF & FARADAY'S LAW]

Step 1. Find the area of the equilateral triangle. From the exam cover: $A = \frac{1}{2}bh$. For an equilateral triangle with side a : base $b = a$, height $h = \frac{\sqrt{3}}{2}a$.

$$A = \frac{1}{2} \cdot a \cdot \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a^2$$

Step 2. Compute flux. The loop is in the y - z plane, so $d\mathbf{s} = \hat{x} ds$.

$$\Phi = \iint_S \mathbf{B} \cdot d\mathbf{s} = \iint B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t) \cdot \hat{x} ds$$

Use DOT PRODUCTS table: $\hat{x} \cdot \hat{x} = 1$, $\hat{y} \cdot \hat{x} = 0$.

$$\Phi = B_0(-2)(1) \sin(\omega t) \iint ds + B_0(3)(0) \sin(\omega t) \iint ds = -2B_0 \sin(\omega t) \cdot A$$

$$\Phi = -\frac{\sqrt{3}}{2} B_0 a^2 \sin(\omega t) \text{ Wb}$$

(b) EMF induced in the loop. (6 pts)

[Suppl: p.1 col 3: $V_{\text{emf}} = -N d\Phi/dt$; $\partial B/\partial t$ TABLE]

Step 1. Differentiate Φ with respect to t .

$$\frac{d\Phi}{dt} = -\frac{\sqrt{3}}{2} B_0 a^2 \omega \cos(\omega t) \quad (\text{from } \partial_t \sin(\omega t) = \omega \cos(\omega t) \text{ in } \partial B/\partial t \text{ table})$$

Step 2. Apply Faraday: $V_{\text{emf}} = -N d\Phi/dt$.

$$V_{\text{emf}} = -N \left(-\frac{\sqrt{3}}{2} B_0 a^2 \omega \cos(\omega t) \right)$$

$$V_{\text{emf}} = \frac{\sqrt{3}}{2} N B_0 a^2 \omega \cos(\omega t) \text{ V}$$

(c) Current through 1 k Ω resistor at given values. (8 pts)

[Suppl: p.1 col 3: Equivalent circuit $I = V_{\text{emf}}/R$; p.1 col 2: UNIT CIRCLE $\cos(\pi/4) = \sqrt{2}/2$]

Plug in $N = 10$, $a = 0.08$ m, $B_0 = 0.02$ T, $\omega = 2 \times 10^6$ rad/s, $\omega t = \pi/4$, $R = 1$ k $\Omega = 10^3$ Ω .

Step 1. Combine $\cos(\pi/4) = \sqrt{2}/2$ with the $\sqrt{3}/2$ factor: $\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{4}$. Then:

$$V_{\text{emf}} = \frac{\sqrt{6}}{4} \cdot N \cdot B_0 \cdot a^2 \cdot \omega = \frac{\sqrt{6}}{4} \cdot 10 \cdot 0.02 \cdot (0.08)^2 \cdot 2 \times 10^6$$

Numeric bracket step by step:

$$(0.08)^2 = 6.4 \times 10^{-3}, \quad 0.02 \times 6.4 \times 10^{-3} = 1.28 \times 10^{-4}, \quad 1.28 \times 10^{-4} \times 2 \times 10^6 = 256, \quad 10 \times 256 = 2560$$

$$V_{\text{emf}} = \frac{\sqrt{6}}{4} \times 2560 = 640\sqrt{6} \approx 1568 \text{ V}$$

Step 2. Since loop resistance is negligible ($R_i = 0$), from EQUIVALENT CIRCUIT:

$$I = \frac{V_{\text{emf}}}{R} = \frac{640\sqrt{6}}{10^3}$$

$$I = 0.64\sqrt{6} \approx 1.57 \text{ A}$$

Problem 2 (12 pts) — Motional EMF, Moving Wire

Given: Wire of length $a = 15$ cm = 0.15 m along $\hat{y}$. Velocity $\mathbf{u} = 12\hat{x}$ cm/s = 0.12 $\hat{x}$ m/s. Field $\mathbf{B} = (0.1\hat{x} - 0.05\hat{z})$ T.

[Suppl: p.1 col 3: Motional EMF $V^m = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$; p.1 bottom: CROSS PROD CYCLE]

Step 1. Compute $\mathbf{u} \times \mathbf{B}$.

$$\mathbf{u} \times \mathbf{B} = 0.12\hat{x} \times (0.1\hat{x} - 0.05\hat{z}) = 0.012 \underbrace{(\hat{x} \times \hat{x})}_{=0} - 0.006 \underbrace{(\hat{x} \times \hat{z})}_{=-\hat{y}} = 0 - 0.006(-\hat{y}) = 0.006\hat{y} \text{ V/m}$$

(From CROSS PROD CYCLE: $\hat{x} \times \hat{z} = -(\hat{z} \times \hat{x}) = -\hat{y}$.)

Step 2. Integrate along the wire. Wire lies along $\hat{y}$, so $d\mathbf{l} = \hat{y} dy$.

$$V_{\text{emf}} = \int_0^{0.15} (0.006\hat{y}) \cdot (\hat{y} dy) = 0.006 \times 0.15$$

$$V_{\text{emf}} = 9 \times 10^{-4} \text{ V} = 0.9 \text{ mV}$$

Note: Only the $-0.05\hat{z}$ component of $\mathbf{B}$ contributes: $\hat{x} \times \hat{x} = 0$ eliminates the $\hat{x}$ component entirely.

Problem 3 (12 pts) — Conduction vs. Displacement Current

Given: $I_c = 2 \sin(\omega t) \mu\text{A} = 2 \times 10^{-6} \sin(\omega t) \text{ A}$ through area A . Distilled water: $\sigma = 10^{-4} \text{ S/m}$, $\epsilon_r = 26\pi$, $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$.

(a) Electric field E . (6 pts)

[Suppl: p.2 col 1: CONDUCTION vs. DISPLACEMENT CURRENT; $J_c = \sigma E$, $I_c = J_c A$]

Step 1. Current density: $J_c = I_c/A$. **Step 2.** Ohm's law: $J_c = \sigma E \Rightarrow E = J_c/\sigma = I_c/(\sigma A)$.

$$E = \frac{2 \times 10^{-6} \sin(\omega t)}{10^{-4} \cdot A}$$

$$E = \frac{0.02}{A} \sin(\omega t) \text{ V/m}$$

(b) Displacement current I_d at $\omega = 10^9 \text{ rad/s}$, $\omega t = \pi/3$. (6 pts)

[Suppl: p.2 col 1: $J_d = \epsilon \partial E / \partial t$, $I_d = J_d A$; UNIT CIRCLE: $\cos(\pi/3) = 1/2$]

Step 1. Compute $\epsilon = \epsilon_r \epsilon_0$.

$$\epsilon = 26\pi \cdot \frac{1}{36\pi} \times 10^{-9} = \frac{26}{36} \times 10^{-9} = \frac{13}{18} \times 10^{-9} \text{ F/m}$$

Step 2. $I_d = \epsilon A \partial E / \partial t = \epsilon A \cdot \frac{0.02}{A} \cdot \omega \cos(\omega t) = \epsilon \cdot 0.02 \cdot \omega \cos(\omega t)$.

At $\omega t = \pi/3$:

$$I_d = \frac{13}{18} \times 10^{-9} \times 0.02 \times 10^9 \times \frac{1}{2} = \frac{13}{18} \times 0.02 \times 0.5 = \frac{13}{1800}$$

$$I_d = \frac{13}{1800} \approx 7.22 \text{ mA}$$

Note: $I_d \approx 7.22 \text{ mA} \gg I_c = 2 \mu\text{A}$. The water stores energy capacitively ($\epsilon_r = 26\pi \approx 82$) much more than it dissipates resistively at GHz.

Problem 4 (10 pts) — Charge Relaxation in Distilled Water

Given: $\sigma \approx 10^{-4} \text{ S/m}$, $\epsilon_r \approx 26\pi$.

(a) Relaxation time constant. (5 pts)

[Suppl: p.2 col 1: CHARGE RELAXATION $\tau_r = \epsilon/\sigma$]

$$\tau_r = \frac{\epsilon_r \epsilon_0}{\sigma} = \frac{\frac{13}{18} \times 10^{-9}}{10^{-4}} = \frac{13}{18} \times 10^{-5} \text{ s}$$

$$\tau_r = \frac{13}{18} \times 10^{-5} \approx 7.22 \mu\text{s}$$

(b) Time for ρ_v to reach e^{-3} of initial value. (5 pts)

[Suppl: p.2 col 1: $\rho_v(t) = \rho_{v0} e^{-t/\tau_r}$]

Step 1. Set up equation:

$$\rho_v(t) = \rho_{v0} e^{-t/\tau_r} = \rho_{v0} e^{-3} \Rightarrow \frac{t}{\tau_r} = 3 \Rightarrow t = 3\tau_r$$

Step 2. Compute:

$$t = 3 \times 7.22 \mu\text{s}$$

$$t = 3\tau_r \approx 21.7 \mu\text{s}$$

Problem 5 (30 pts) — Lossless Plane Wave

Given: Nonconducting ($\sigma = 0$) medium, $\varepsilon_r = 6$, $\mu_r = 1$.

$$\mathbf{E}(z, t) = 4 \cos\left(6\pi \times 10^8 t - kz - \frac{\pi}{6}\right) \hat{y} \text{ V/m}$$

(a) Find ω , k , λ , η .

(10 pts)

[Suppl: p.2 bottom: WAVE PARAMS; $k = \omega\sqrt{\mu\varepsilon}$, $\eta = \sqrt{\mu/\varepsilon}$, $\lambda = 2\pi/k$]

Read ω directly:

$$\omega = 6\pi \times 10^8 \text{ rad/s}$$

Compute k :

$$k = \omega\sqrt{\mu_r\mu_0\varepsilon_r\varepsilon_0} = \frac{\omega}{c}\sqrt{\mu_r\varepsilon_r} = \frac{6\pi \times 10^8}{3 \times 10^8} \sqrt{1 \times 6} = 2\pi\sqrt{6}$$

$$k = 2\pi\sqrt{6} \approx 15.4 \text{ rad/m}$$

Compute λ :

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{2\pi\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$$\lambda = \frac{1}{\sqrt{6}} \approx 0.408 \text{ m}$$

Compute η :

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \eta_0\sqrt{\frac{\mu_r}{\varepsilon_r}} = 120\pi\sqrt{\frac{1}{6}} = \frac{120\pi}{\sqrt{6}} = 20\pi\sqrt{6}$$

$$\eta = 20\pi\sqrt{6} \approx 154 \Omega$$

(b) Find $\tilde{\mathbf{E}}(z)$.

(5 pts)

[Suppl: p.2 col 2: PHASOR METHOD step 2: $\cos(\omega t - kz + \phi) \rightarrow e^{j\phi}e^{-jkz}$]

Strip the $e^{j\omega t}$ factor. From $\mathbf{E}(z, t) = 4 \cos(\omega t - kz - \pi/6)\hat{y}$:

$$\cos(\omega t - kz - \pi/6) \longleftrightarrow e^{-j\pi/6}e^{-jkz}$$

$$\tilde{\mathbf{E}}(z) = 4 e^{-j\pi/6} e^{-jkz} \hat{y} = 4 e^{-j(kz+\pi/6)} \hat{y} \text{ V/m}$$

(with $k = 2\pi\sqrt{6} \text{ rad/m}$)

(c) Find $\tilde{\mathbf{H}}(z)$. (5 pts)

[Suppl: p.2 col 2: DIRECTION RELATION $\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}}$; table row: $\hat{k} = +\hat{z}$, $\hat{E} = +\hat{y} \Rightarrow \hat{H} = \hat{z} \times \hat{y} = -\hat{x}$]

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}} = \frac{1}{\eta} \hat{z} \times (4e^{-j(kz+\pi/6)} \hat{y}) = \frac{4}{\eta} e^{-j(kz+\pi/6)} (\hat{z} \times \hat{y}) = \frac{4}{\eta} e^{-j(kz+\pi/6)} (-\hat{x})$$

With $\eta = 20\pi\sqrt{6}$: $\frac{4}{\eta} = \frac{4}{20\pi\sqrt{6}} = \frac{1}{5\pi\sqrt{6}}$.

$$\tilde{\mathbf{H}}(z) = -\frac{1}{5\pi\sqrt{6}} e^{-j(kz+\pi/6)} \hat{x} \text{ A/m}$$

(d) Find $\mathbf{H}(z, t)$. (5 pts)

[Suppl: p.2 col 2: step 4: $\mathbf{H} = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$]

$$\mathbf{H}(z, t) = \text{Re}\left\{-\frac{1}{5\pi\sqrt{6}} e^{-j(kz+\pi/6)} e^{j\omega t} \hat{x}\right\}$$

$$\mathbf{H}(z, t) = -\frac{1}{5\pi\sqrt{6}} \cos\left(6\pi \times 10^8 t - 2\pi\sqrt{6} z - \frac{\pi}{6}\right) \hat{x} \text{ A/m}$$

(e) Average power density. (5 pts)

[Suppl: p.2 col 3: POYNTING & POWER (lossless): $\mathbf{S}_{av} = |E_0|^2 / (2\eta) \hat{k}$]

$$\mathbf{S}_{av} = \frac{|E_0|^2}{2\eta} \hat{k} = \frac{4^2}{2 \times 20\pi\sqrt{6}} \hat{z} = \frac{16}{40\pi\sqrt{6}} \hat{z} = \frac{2}{5\pi\sqrt{6}} \hat{z}$$

$$\text{Simplify: } \frac{2}{5\pi\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{2\sqrt{6}}{30\pi} = \frac{\sqrt{6}}{15\pi}.$$

$$\mathbf{S}_{av} = \frac{\sqrt{6}}{15\pi} \hat{z} \approx 0.052 \hat{z} \text{ W/m}^2$$

Problem 6 (16 pts) — Lossy EM Wave in Distilled Water

Given: $f = \frac{2}{5} \text{ GHz} = 0.4 \times 10^9 \text{ Hz}$, $\sigma \approx 10^{-4} \text{ S/m}$, $\varepsilon_r \approx 26\pi$, $\mu_r = 1$, wave in $+\hat{z}$, $|E(0)| = 4 \text{ V/m}$.

Useful intermediate values (compute once, use everywhere):

$$\omega = 2\pi f = 2\pi \times 4 \times 10^8 = 8\pi \times 10^8 \text{ rad/s}$$

$$\varepsilon' = \varepsilon_r \varepsilon_0 = 26\pi \cdot \frac{1}{36\pi} \times 10^{-9} = \frac{13}{18} \times 10^{-9} \approx 7.22 \times 10^{-10} \text{ F/m}$$

(a) Find $\varepsilon''/\varepsilon'$. (6 pts)

[Suppl: p.2 col 3: MEDIUM TYPES; $\varepsilon'' = \sigma/\omega$, $\varepsilon' = \varepsilon_r \varepsilon_0$]

$$\varepsilon'' = \frac{\sigma}{\omega} = \frac{10^{-4}}{8\pi \times 10^8} = \frac{1}{8\pi} \times 10^{-12} \text{ F/m}$$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega \varepsilon_r \varepsilon_0} = \frac{10^{-4}}{8\pi \times 10^8 \times \frac{13}{18} \times 10^{-9}} = \frac{10^{-4}}{\frac{104\pi}{18} \times 10^{-1}} = \frac{10^{-4} \times 18}{104\pi \times 10^{-1}} = \frac{18}{104\pi} \times 10^{-3} = \frac{9}{52\pi} \times 10^{-3}$$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{9}{52\pi} \times 10^{-3} \approx 5.5 \times 10^{-5}$$

(b) What type of medium is distilled water at this frequency? (5 pts)

[Suppl: p.2 col 3: MEDIUM TYPES table; threshold $\varepsilon''/\varepsilon' \ll 1$ (i.e. < 0.01)]

$$\frac{\varepsilon''}{\varepsilon'} \approx 5.5 \times 10^{-5} \ll 1$$

Low-loss dielectric $\left(\frac{\varepsilon''}{\varepsilon'} \ll 1 \right)$

Justification: the ratio is $\approx 10^{-4}$, far below the 10^{-2} threshold for quasi-conductor and far below 100 for good conductor.

(c) Find the skin depth. (5 pts)

[Suppl: p.2 col 3: MEDIUM TYPES, low-loss row: $\alpha \approx \frac{\sigma}{2} \sqrt{\mu/\varepsilon'}$; $\delta_s = 1/\alpha$]

Step 1. Compute $\eta' = \sqrt{\mu_0/\varepsilon'}$ (intrinsic impedance).

$$\eta' = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{26\pi}} \approx \frac{376.99}{9.04} \approx 41.7 \Omega$$

Step 2. Apply low-loss approximation $\alpha \approx (\sigma/2) \eta'$:

$$\alpha \approx \frac{\sigma}{2} \cdot \eta' = \frac{10^{-4}}{2} \times 41.7 = 2.085 \times 10^{-3} \text{ Np/m}$$

Step 3. Skin depth $\delta_s = 1/\alpha$:

$$\delta_s = \frac{1}{2.085 \times 10^{-3}}$$

$\delta_s \approx 480 \text{ m}$

Note: 480 m skin depth confirms this is a near-lossless medium at 0.4 GHz with $\sigma = 10^{-4} \text{ S/m}$. The wave barely attenuates — it can travel hundreds of meters before losing a factor of e .

Blue labels [Suppl: ...] point to the exact section of the Exam 1 Supplement to look up.